

Section 80 — Undirected Connectivity Decomposition

Low links describe escape routes

This lesson derives Problems A–C. Problem D is reserved for the editorial.

Problem A: bridges and the bridge forest

During DFS, `tin[u]` is the entry time and `low[u]` is the smallest entry time reachable from `u`'s subtree using tree edges and at most one back edge. A tree edge `u-v` is a bridge exactly when `low[v] > tin[u]`: the subtree has no route to `u` or an ancestor without that edge.

Remove bridges and flood-fill the remaining edge-biconnected components. Every bridge becomes an edge between components, forming a tree for a connected graph. Therefore the mandatory bridge count between two vertices is their tree distance, answered by LCA.

Use edge IDs, not parent vertices: otherwise two parallel edges are incorrectly treated as one bridge.

Problem B: articulation damage

For a child `v` with `low[v] >= tin[u]`, deleting `u` isolates the entire DFS subtree of `v`. These child-subtree sizes and the one remaining outside group partition all surviving vertices. The answer is the sum of products over distinct groups; accumulate it incrementally as `answer += seen * group_size`.

Problem C: block-cut trees

Keep DFS edges on a stack. Whenever `low[v] >= tin[u]`, pop through edge `u-v`; the popped endpoints form one vertex-biconnected block. Build a bipartite tree containing original vertices and block nodes. An original vertex `c` lies on every `u-v` path exactly when its node lies on the unique block-cut-tree path from `u` to `v`, testable with three distances.

Worked traces and implementation

If DFS uses one of two parallel $u-v$ edges as its tree edge, the other is a back edge to u , so $\text{low}[v] \leq \text{tin}[u]$; neither is a bridge. Skip only the exact parent edge ID, not every edge whose endpoint is the parent.

If deleting u leaves group sizes 2,3,4, separated pairs number $2*3+2*4+3*4$. Incrementally add $\text{seen}*\text{group_size}$ and include the outside group. The DFS root is special: it has no ancestor-side group and is an articulation vertex only with at least two DFS children.

Block nodes and original vertices form a tree because a cycle in their incidence graph would join its blocks into a larger biconnected block. Thus path membership is tested by $\text{dist}(u,c)+\text{dist}(c,v)=\text{dist}(u,v)$ after endpoint cases are handled.

Use shared edge IDs and an edge stack for block extraction. Tiny oracles should delete each edge or vertex and rerun BFS; include parallel edges, a star, one cycle, and a DFS root with one child.

Exercises

A–C develop bridges, articulation partitions, and block-cut paths. D is the independent transfer problem.